observing the progress of healing bones. Further, in a finished metal product, it is used to detect faults, cracks, flaws and holes.

Gamma rays

It is produced by transitions of radioactive nuclei and decay of certain elementary particles. They produce chemical reactions on photographic plates, fluorescence, ionisation, diffraction. The frequency range is 10^{18} Hz and above.

Gamma rays have higher penetrating power than X-rays and ultraviolet radiations; it has no charge but harmful to human body. Gamma rays provide information about the structure of atomic nuclei. It is used in radio therapy for the treatment of cancer and tumour, in food industry to kill pathogenic microorganism.

EXAMPLE 5.4

A magnetron in a microwave oven emits electromagnetic waves (em waves) with frequency f = 2450 MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?.

Solution

Frequency of the electromagnetic waves given, f = 2450 MHz

The corresponding angular frequency is

$$\omega = 2\pi f = 2 \times 3.14 \times 2450 \times 10^6$$
$$= 15{,}386 \times 10^6 \text{ Hz}$$
$$= 1.54 \times 10^{10} \text{ s}^{-1}$$

The required magnetic field, $B = \dfrac{m_e \omega}{|q|}$

Mass of the electron, $m_e = 9.11 \times 10^{-31}$ kg

Charge of the electron,

$$q = -1.60 \times 10^{-19} \text{C}$$
$$\Rightarrow |q| = 1.60 \times 10^{-19} \text{C}$$

$$B = \frac{\left(9.11 \times 10^{-31}\right)\left(1.54 \times 10^{10}\right)}{\left(1.60 \times 10^{-19}\right)} = 8.7683 \times 10^{-2} \text{T}$$

$$B = 0.08768 \text{ T}$$

This magnetic field can be easily produced with a permanent magnet. So, electromagnetic waves of frequency 2450 MHz can be used for heating and cooking food because they are strongly absorbed by water molecules.

5.3 TYPES OF SPECTRUM- EMISSION AND ABSORPTION SPECTRUM- FRAUNHOFER LINES

When an object burns, it emits radiations. That is, it emits electromagnetic radiation which depends on temperature. If the object becomes hot, it glows in red colour. If the temperature of the object is further increased, then it glows in reddish-orange colour and becomes white when it is hottest. The spectrum in Figure 5.11 usually

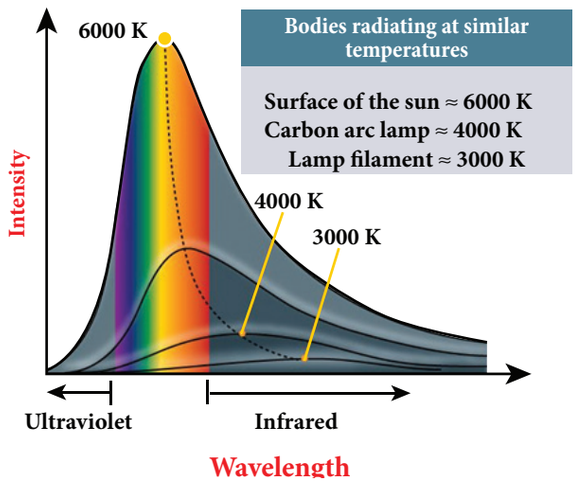


Figure 5.11 Black body radiation spectrum – variation with temperature

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